

Machine Learning and its Applications

Today's goals

Module 1

Machine Learning Fundamentals I: Basic Concepts

Welcome

Welcome to the specialization - Applied AI for Engineers and Scientists: Foundations

Video • 6 min

Specialization and Course Structure

Reading • 10 min

Specialization Sample Certificate

Reading • 10 min

Stay connected with UofG Online

Reading • 5 min

Module 7 Introduction

Introduction to Module 7's Study

Video • 1 min

How to Access MATLAB Online

Reading • 5 min

Machine Learning in Engineering Practice: Tool, Replacement, or Decision Aid?

Reading • 5 min

Module 7-1 Machine Learning Fundamentals

Machine Learning Fundamentals: What is Machine Learning

Video • 6 min

Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (1)

Video • 6 min

Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (2)

Video • 8 min

Materials for Machine Learning Fundamentals

Reading • 10 min

From Explicit Rules to Learned Models: What Actually Changes?

Reading • 5 min

Module 1 Learning Coach

Dialogue • 5 min

Module 7-2 Mapping Inputs to Outputs

Mapping Inputs to Outputs: Data Representation

Video • 10 min

Mapping Inputs to Outputs: Parametric ML Model

Video • 7 min

Mapping Inputs to Outputs: Non-Parametric ML Model

Video • 7 min

Mapping Inputs to Outputs: Evaluate Output

Video • 2 min

Materials for Mapping Inputs to Outputs

Reading • 10 min

How Data Representation and Model Choice Shape What a Model Can Learn

Reading • 5 min

Module 7-3 MATLAB Implementation

MATLAB Implementation: Simple Linear Regression and KNN

Video • 6 min

Materials about MATLAB Implementation

Reading • 10 min

From Concept to Code: What the Model Is Actually Doing

Reading • 5 min

Module 7 Quiz

Module 7 Quiz

Practice Assignment • 30 min

Module 7 Assignments

Module 7 Assignment 1

Graded App Item • 30 min

Module 7 Assignment 2

Graded App Item • 30 min

CSV File for Module 7 Assignment

Reading • 1 min

Module 7 Recap

Reading • 5 min

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 3

Data Preparation

Module 4

Support Vector Machines

Module 5

Artificial Neural Networks

From Concept to Code: What the Model Is Actually Doing



This module introduced MATLAB implementations of **linear regression** and **k-nearest neighbors**, using real datasets.

For **linear regression**:

- What assumption about the relationship between input and output is being made?
- Why does adding an intercept term change the fitted model, and in what situations would omitting it be inappropriate?

For **KNN**:

- Why does the choice of k affect prediction results?
- In what way is KNN fundamentally different from linear regression in how it uses data at prediction time?

In either method, explain **one scenario** where the MATLAB code would still run correctly, but the *model's output would be misleading*.

Go to next item

✓ Completed

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